

Logan Barnett's Résumé

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Location Beaverton, Oregon, USA (remote)

Summary

22 years as a professional software engineer, spanning video games, embedded systems, web applications, and the tooling that engineering and operations teams depend on. My current focus is systems programming in Rust, building reproducible tooling with NixOS and nixpkgs, and applying LLMs as deliberate engineering tools — including a practical understanding of where they fail and how to design workflows that account for that. I care about sustainable software: strong type systems, thorough testing, and documentation that makes code legible to the next engineer. I enjoy mentoring and gravitate toward the hard problems.

Job Experience

Houghton Mifflin Harcourt (HMH, formerly NWEA) (2023–current)

Embedded software engineer on the DevOps team — the person who builds the tools the team runs on, rather than running them. The team manages over 2000 VMs via Puppet and r10k; my role is the software layer above that: deployment orchestration, environment management tooling, and the internal systems that let the team operate at scale without coordinating by hand.

1. Took ownership of Environment Down Days (EDD) — a recurring operational process that had grown into organized chaos: 10 engineers working in parallel for 3 days, each with their own bespoke scripts. Recognizing that EDD and the existing deployment tooling were solving the same problem twice, I unified them, paid down significant tech debt, and introduced parallelism where it was safe. Result: 10 engineers $\times$ 3 days collapsed to 1 engineer $\times$ 1–1.5 days, with a system that is correct and maintainable rather than just fast.
2. Mentored 4–6 engineers at a time on the Puppet ecosystem, deployment workflows, and automation practices.
3. Migrated the primary internal tooling suite (nwea-gem) and extracted libraries from it to CICD, and converted approximately a dozen Jenkins freestyle jobs to Groovy-based Jenkinsfiles.
4. Used LLMs extensively for troubleshooting, code edits, and automating repetitive engineering tasks via CLI.

Principal Instructor at Alchemy Code Lab (2022–2023)

Taught 3–4 cohorts of roughly 20 students each across a year at a code school that had transitioned permanently to remote delivery following Covid.

1. Taught advanced JavaScript topics: Promises and async patterns, type safety, reducers, and Data Structures and Algorithms.
2. Stressed simplicity over ease — a meaningful distinction when teaching React. A recurring focus was guiding students to recognize when to step back, refactor, and reach for reducers or functional patterns instead of defaulting to `useState`.

3. Mentored students individually on software engineering practices beyond the curriculum, including how to think about maintainability and code that the next person can understand.

NWEA (2016–2022)

Hired for JavaScript expertise; spent the first several years on frontend and application engineering before transitioning to DevOps at the onset of the COVID lockdown in 2020.

DevOps (~2020–2022)

Joined entirely remote — interviewed on-site but never worked there in person. Brought a software engineer’s perspective on what it actually feels like to consume DevOps tooling, which shaped a lot of the improvements made.

1. Assumed stewardship of the primary internal Ruby tooling suite (nweagem), which controlled colo management, deployments, release planning, and credential synchronization. Improved reproducibility, test coverage, memoized API calls, and automated deployments to keep the suite functional in a rapidly changing environment.
2. Directed the Build Engineering sub-team’s effort to consolidate the build infrastructure: from ~8 largely unmanaged VMs (engineers installing tools manually as needed) to the Cloud Swarm — a flexible pool of Dockerized Jenkins agents under proper configuration management.
3. Maintained and extended the Nomad scheduling infrastructure that ran in-house deployment and management tools (Kraken, Compass) alongside approximately a dozen engineer-built applications.

Frontend and Application Engineering (~2016–2020)

Senior engineer leading the build of a new React-Redux proctoring interface for running student tests at scale — approximately 100,000–200,000 concurrent users at peak, where AWS API limits were a genuine constraint.

1. Achieved 100% Flow type coverage across the UI, treating the type system as a first-class testing tool rather than an afterthought.

2. Built a suite of 350+ Cucumber + Nightwatch.js acceptance tests that QA members could author and maintain without deep engineering knowledge. The suite ran in ~50 minutes on feature branches; a `--ff-only` merge policy meant master was bit-for-bit identical to the branch and did not need to be retested.
3. Built a feature flag system from scratch — deliberately simple, local to the UI, and fully integrated with the test suite. Tests declared which flags they required via Cucumber tags, so every feature launched with "flag off" tests already in place before "flag on" tests were added.
4. Kept API mocks automatically synchronized with Swagger definitions, making the UI a reliable enforcer of API contracts and catching back-end drift before it reached production.

UTi Worldwide / DSV (via IT-Motives) (2014–2016)

Hired specifically as AngularJS expertise to lead the UI of a customer-facing shipment tracking portal at a global supply chain coordinator. Tenure ended when DSV acquired UTi and shut down the Portland office.

1. Restructured the AngularJS application architecture to support a growing UI surface area; established TDD on the team and brought the UI to 100% test coverage.
2. Set up Jenkins for automated CI with test coverage reporting; introduced Cucumber for executable business requirements.
3. Led a React + Redux pilot project (self-service identity management against Oracle OIM/OID) intended to demonstrate a path for modernizing legacy applications incrementally.
4. Served as informal AngularJS consultant to other UTi teams evaluating the framework.

Early Career (2004–2014)

Software engineering roles spanning educational games, hosted services, Rails consulting, embedded device tooling, and enterprise .NET:

- *Arizona State University* → *E-Line Media (2011–2014)* — Co-lead and senior developer on AtlantisRemixed, an educational game built for academic research on the efficacy of video games as a teaching medium.

Led a team of 4 engineers; built a 2D visual scripting tool for designers; designed the build and patching systems in Unity / .NET / Mono; integrated with a Ruby on Rails backend.

- *GoDaddy (2010–2011)* — Hosted Exchange: migrated management from the Outlook API to PowerShell; built internal support tooling in MVC3; extensive PowerShell work against Exchange 2010.
- *Integrum Technologies (2009–2010)* — Rails consulting shop. Pair-programmed across client projects including fantasy sports backends, a legacy Rails app with thousands of users, and a public transit schedule system.
- *Happy Camper Studios (2007–2009)* — Built configuration and diagnostics desktop applications for satellite modems (Radyne / Comtech). The scale of the UI directly drove the creation of Monkeybars, a JRuby-based MVC framework for Swing applications, which was later generalized and open-sourced.
- *UHaul International (2004–2007)* — First engineering position out of DeVry. .NET desktop applications, SOAP web services, .NET remoting, and a Java/.NET integration layer for a self-insurance claims system.

Full details available in the long-form CV.

Proficiencies

Languages

Javascript

React, Redux, Three, Angular, Node, ES6, Webpack, Flow, npm, yarn.

Rust

Actix, futures, abstract data types, monad chains, and avoiding `unwrap` ;)

Nix

NixOS, nixpkgs, home-manager, nix-darwin, flakes, declarative system configuration, upstream nixpkgs contributions.

Ruby

Ruby on Rails, Sinatra, JRuby, Cucumber, RSpec.

Java

Swing, JAX-RS, JRuby, JUnit, SNMP4J.

.net

C#, Boo, Unity/Mono, MVC, WCF, NUnit, OData, Powershell.

Infrastructure and Ops

Puppet (Hiera, r10k; enough to extend and maintain manifests), Docker, Nomad, NATS, Jenkins.

Databases

PostgreSQL, MongoDB, MySQL, SQL Server, Oracle + PL/SQL.

LLMs

Agentic workflows, MCP (Model Context Protocol), prompt engineering, tool use, Claude, automating mundane and non-trivial engineering tasks.

Misc

JIRA, Pivotal Tracker, Trello, git, Perforce, svn, hg, Plastic SCM, vim, emacs, literate programming, functional programming.

Selected Side Projects

1. dns-smart-block — A Rust workspace of four co-orchestrating programs that bring LLM-powered classification to DNS-level content blocking. A log processor watches DNS queries and enqueues domains via NATS; a queue processor dispatches them to a classifier; the classifier uses a locally-hosted LLM (Ollama) as a fuzzy matching algorithm — returning a classification, confidence score, and reasoning; a block-list server exposes the results in a format compatible with Pi-hole, Ad-Guard Home, Blocky, and others. Event-sourced PostgreSQL storage,

Prometheus metrics, health checks, and an admin web view. Deployed via NixOS flake.

2. garage-queue — A capability-aware work queue written in Rust. Producers submit payloads; the queue routes each item to a worker that advertises the right hardware and software capabilities to handle it — the queue itself has no knowledge of any particular workload, since routing logic lives entirely in configuration. Serves as the central dispatcher for a heterogeneous fleet of inference machines, routing Ollama requests to hosts with sufficient VRAM and the right models loaded.
3. sonify-health — Turns infrastructure health into ambient sound. Each machine self-reports via percussive heartbeat boops and a continuous stress drone — a Star Trek bridge feel that encodes real meaning into peripheral awareness without requiring visual attention. Written in Rust.
4. nix-config — Nix configuration for an entire self-managed home network. Every host is defined as code and deployed with a custom copy-closure tool — the foundation that everything else (DNS, identity, declarative API management) builds on.
5. sytter — Event-driven host automation: IFTTT for your machine. Written in Rust. Reads TOML-defined pipelines of triggers (system/kernel events), conditions (stateful guards), and executors (actions). Designed to run on macOS, Linux, and Windows.
6. Lambda Cast — Co-hosted ~20 episodes on functional programming concepts (2016–2019). Episodes still available.

Education

DeVry University

Graduated 2004 with BS in Computer Engineering Technology.